

Institute of Actuaries of India

Subject SA4 – Pensions and Other Benefits

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) The Company X may want:**

- 1) To assess the financial strength of scheme Y and the adequacy of its fund to meet the accrued benefits, both on “going concern” basis and “closed” basis [0.50]
 - 2) To address the issue of inadequacy of the fund of scheme Y, if any before finalization of purchase price (either by immediate injection of funds by company Y or through an adjustment in purchase price) [0.50]
 - 3) To assess the annual cost of benefits of scheme Y and its continued viability [0.50]
 - 4) To understand the other issues that may crop up following the merger which may have financial consequences for Company X (eg demand from employees of Company X who are now covered under a DC scheme to be included under the DB scheme) [0.50]
 - 5) To address the concern of the trustees of scheme X who may like
 - that the scheme Y be funded adequately before merger of schemes in order to protect the security of benefits of members of scheme X [0.50]
 - To compare the valuation methodology and basis of scheme Y with that of scheme X [0.25]
 - To know whether assets of scheme Y are valued at book value or market value or some other value [0.25]
 - To know about the quality of assets of scheme Y [0.25]
- [Max 3]**

ii) I may be interested to look into the following aspects of scheme Y with their comparison to scheme X:

- a) Nature & level of benefits of scheme Y and their comparison with those of scheme X
 - The nature & level of benefits of scheme Y such as eligibility conditions; accrual formula; pensionable salary; pension increases- whether discretionary or guaranteed, if any; spouse/family pension; commutation terms; vesting conditions etc. [0.50]
 - The comparative benefits of DB scheme X and similarity/dis-similarities [0.25]
 - Accrual may be 1% or 1/66th etc. for each year of service; pensionable salary may be last drawn, or average of last 12 months or last 36 months or even career average; similarly vesting conditions may require 5,10 or 20-years’ service etc. [0.50]
 - Scheme Rules of both the schemes X & Y need to be looked into including their amendments, if any for this purpose [0.25]

[Max 1.5]
- b) Profile of members of scheme Y:
 - Besides the level & nature of benefits, the profile of members such as number of members, their ages, pensionable salary, increases in salary, attrition rates, past service etc. may help to assess the ongoing pension cost of the scheme [0.50]
 - The movement of profile of members i. e. how it changes over the period is also important. An increasing number of members, increasing age profile and increasing salary level may increase pension cost or vice versa [0.50]
 - Similarly if vesting conditions are onerous, higher attrition rate may show declining cost or vice versa [0.25]
 - A stable & mature profile of members will show stable cost of scheme whereas a changing profile will yield volatility of pension cost [0.25]
 - A large influx of new entrants especially at higher ages may affect the pension cost significantly [0.25]

[Max 1.5]

c) Pension cost & its volatility;

- The true pension cost would depend on the nature & level of benefits; the profile of employees and actual experience in respect of salary increases, attrition etc. as reflected over time [0.50]
 - Estimation of cost is made using a valuation method and a set of assumptions which may vary over time. Valuation method may depend on the purpose of valuation- whether funding or accounting of pension costs etc. The assumptions are based on long term expected future experience by members covered [0.75]
 - While the true cost is certain but is not known in advance and hence the same is estimated using a basis. The estimated cost may vary from one year to another depending on the actual experience being quite different from expected. The estimated cost may also vary due to change of basis [0.75]
 - Since the scheme is funded, the cost of scheme will also depend on the investment performance of the funds of the scheme. [0.25]
- [Max 2]

d) Funding status of scheme Y:

- Will help to assess the covenant of Company Y to finance the pension scheme Y. While small deficit is understandable, a huge/increasing deficit reflects the inability/financial constraints of Company Y [0.50]
 - Large and continuing deficits offer poor security of the vested benefits & threaten the viability of the Scheme [0.25]
 - Funding status will depend on the value of liabilities estimated and the value of assets held by the scheme which might have been valued as book value or market value or some other value. [0.50]
 - The variation of method and/or basis for valuation of liabilities and assets may lead to volatility of funding status [0.25]
 - Past actuarial valuation reports may reveal the method, basis, strength & inherent risk characteristics of the scheme Y [0.25]
- [Max 1.5]

e) Investment & quality of assets:

- Inventory of assets will reveal the nature & level of the investments held by the scheme- whether G-Sec, Corporate Bonds, Equity, Mutual funds, Money Market instruments etc. [0.50]
 - It will also help to assess the expected return (eg coupon/dividend or other regular income [0.25]
 - The book value and market value of these investments held will give idea of the extent of capital gains anticipated) under the scheme [0.25]
 - This will also reveal the extent of non-performing assets (NPA) & the Self investment (Bonds/Equity shares of Company Y owned by scheme Y) [0.25]
 - This will help to understand the investment strategy of scheme Y; assess to what extent the assets & Liabilities are matched [0.25]
- [Max 1.5]

f) Other areas:

- The interest, reaction and power of other stakeholders such as employee unions, Trustees of scheme Y, IT authorities etc. and their likely response to proposed merger may be important and needs to be considered [0.50]
- [Max 8]**

- iii) Definition (source: para 7 “Definitions” Ind AS 19): Actuarial gains and losses comprise:
- experience adjustments (the effects of differences between the previous actuarial assumptions and what has actually occurred); and
 - the effects of changes in actuarial assumptions.

(0.5 for each max 1)

Treatment of Actuarial gains/losses as per Ind As 19 (source Para 92 & 93 Ind As 19):

- An entity shall recognize immediately in “Other Comprehensive Income” (OCI) all of its actuarial gains and losses [0.50]
- Actuarial gains and losses recognized in other comprehensive income shall be presented in the statement of profit and loss [0.50]
- Actuarial gains and losses that have been recognized in other comprehensive income shall be recognized immediately in retained earnings [0.50]
- They shall not be reclassified to profit or loss in a subsequent period. [0.50]

[Max 3]

- iv) a) Actuarial losses of company Y have been substantial and positive for all the last 5 years [0.50]
- They have grown by more than 211% in the last year [0.25]
- As a % of pension cost they have grown every year during last 5 years from 12% to 40% ([0.50]
- If accounts of company Y have been prepared as per AS15 (R), then actuarial gains/losses would have been recognized in P&L [0.25]
- Significant actuarial gains/losses would lead to volatility of profits [0.25]
- As they have been recognized in the OCI as per Ind AS 19 (assuming IND AS 19 applies), the pension cost as reflected in the P&L account is therefore not affected to that extent [0.50]
- However, OCI for a year is a part of “net earnings” carried to the balance sheet at the year end [0.25]
- The “large” actuarial losses off set the earnings generated from business activities in the financial year [0.25]
- If there are losses continuously over a number of years and have even grown, it may erode the retained earnings of the previous years of the company Y [0.50]
- Further, it also shows that assumptions are far from reality. It may question on credibility of the actuarial valuation and hence the liabilities assessed may not represent true financial position of the company [0.50]
- This would severely impact the dividend declaration policy of company Y & may be adversely viewed by the investors/potential investors [0.50]
- The impact depends upon how significant the actuarial losses are as compared to
- pension cost
 - other components of OCI (e. g. gains/losses of currency fluctuations, derivative instruments)
 - overall Net earnings generated.
- A small, random actuarial gain/loss (gain in some years and loss in some other years) of a pension fund is unlikely to have an impact on the earning capacity of company Y [0.50]
- [Max 5]**
- b) The primary sources of actuarial gains/losses may be:
- Effect of change of assumptions, [0.25]
 - Differences between what was valued & what is paid during the period in case of exits, [0.50]

c) Gains/Losses triggered due to One-off events,	[0.50]
d) Effect of differences arising due to actual experience being different from that expected	[0.25]
e) Lack of in-depth analysis of such gains/losses leading to improper classifications as actuarial gain/loss	[0.50]
Change in valuation assumptions may be in respect of financial assumptions such as discount rate, salary escalation rate, pension increases, if any and demographic assumptions such as mortality (in-service & retirement including spouse), employee attrition rate (including disability & early retirement), proportion of spouses/family members eligible for benefits, age difference between member and spouse, proportion of members opting for commuted pension etc.	[1]
An analysis of actuarial gains/losses arising over two successive accounting period valuations may help to identify the factors contributing them.	[0.50]
Such an analysis is to be carried out separately for assets & liabilities.	[0.50]
An analysis of the difference between actual investment return and the investment returns expected on the assets at the beginning of the year will contribute the gains/losses from the assets side.	[0.50]
Following factors may be causing actuarial gains/losses on the assets side:	
Asset/liability mismatch due to which some assets remain idle or are sold at a loss,	[0.50]
Poor/efficient investment/reinvestment strategy,	[0.25]
Delayed/fast investment decisions,	[0.25]
Faster/delayed cashflows in to/out of scheme funds by way of contribution, benefit payout & expenses	[0.50]
Higher/Lower NPAs than expected,	[0.25]
Higher/lower investment expenses	[0.25]
Similar analysis into the value of liability movements over the successive period may help to identify the following factors causing actuarial losses on the liability side:	
Increase of pensionable salary being different from that assumed	[0.25]
Pension increases different from those assumed	[0.25]
Attrition rate different from that assumed	[0.25]
In-service and /or retirement mortality different from that assumed	[0.25]
Change in proportion of spouse/family members from that assumed	[0.25]
Age difference between members and their spouses being different from assumed	[0.25]
Proportion of members opting for commuted pension different from that assumed	[0.25]
There may be a number of one-off events giving rise to actuarial gains/losses as under:	
Unreconciled differences in data supplied either at last valuation or at current valuation may lead to actuarial gains/losses	[0.50]
Sudden influx of new members into the scheme particularly at higher ages may be a cause of actuarial losses	[0.50]
Unexpected discretionary increases in pension benefits during the period may be a factor of actuarial losses	[0.50]

A comparison of the benefits paid/payable to/in respect of the outgoing members/beneficiaries and the reserve held in the valuation liability for them will reveal the amount of actuarial gains/losses in their respect. [0.50]

This may be due to a number of factors such as:

- Actual experience in respect of a number of factors such as mortality, pensionable salary, attrition (before fulfilling and after fulfilling vesting conditions), pension increases etc. is different from that expected for outgoing members [0.50]
- If actuarial gain (or loss) occur continuously for a number of years, it may indicate that the actuarial assumptions are far away from reality and need to be looked into [0.50]
- The benefits calculated and paid in respect of members/beneficiaries are erroneous indicating a deficiency in the operating system [0.50]

An in-depth analysis of actuarial gains/losses may reveal that the accrued benefits of members have decreased/increased during the accounting period either due to Plan amendment/curtailment/settlement etc. not properly classified as past service cost and wrongly included in actuarial gains/losses. [1]

Factors external to the scheme such as changes in the tax regime, labour laws, not captured in the valuation results properly may also be a source of actuarial gains/losses. [1]

Other factors contributing to actuarial gains/losses may include:

- Approximations used in calculating liability, CSC, interest cost etc. [0.50]
- Approximations used in the timing of cash-flows (both benefit payment, contribution, investment income etc.) [0.50]
- Change in the Actuarial Method [0.25]
- Error in valuation exercise [0.25]

[Max 14]

v) A realistic valuation of the pension liabilities as on the purchase date needs to be carried out. [0.25]

It is necessary to have full members' data for this purpose. Members' data need to be reconciled with Scheme data of last valuation with movements during the period elapsed since then as well as with the financials of the company Y for pensionable salary and the financials of the scheme Y for pension payouts. [0.50]

The benefits valued should be consistent with the scheme benefits and for it trust deed & scheme rules need to be referred. Scheme booklet, if any distributed to members also need to be referred to and it is ensured that there are no differences in benefits between two documents. Judgement is also required when deciding to what extent discretionary benefits, if any are to be taken into account while projecting. For it, history of such benefits granted in the past with their basis, if any is also obtained. [1]

Cash flow model projects the accrual of benefits & the payouts for each member under several contingencies till their working life and after vesting of benefits till the expected life time of members and discounts the net cash flows to the purchase date. [0.50]

The parameters used in the projection model should reflect the best estimate in respect of financial assumptions such as pensionable salary increases, pension increases, investment return on plan assets & demographic assumptions such as in-service & post retirement mortality, attrition, proportion of spouses/family members, age differences between members and spouses, proportion of members opting for commuted pension etc.

[1]

Risk pricing models capture several risks to which the scheme Y is exposed to and longevity is one such risk which is important. Other risks such as salary increases, attrition, operational risks etc. after purchase of company Y will be under control of company X and not under control of company Y. Hence realistic valuation may allow some reasonable risk margin in longevity and may not consider margin under other factors which may be under control of company X.

[1]

The net cash flows over each year are to be discounted using “risk free bond yield” to the present time to arrive at the value of expected pension payouts.

[0.50]

Further, financial assumptions should be consistent with each other.

[0.25]

The value of payouts less the market value of the assets held at the purchase date of company Y gives the net realistic value of liabilities of the pension scheme Y.

[0.50]

It is usual for the actuary when arriving at the actuarial value of liabilities to estimate sensitivity results to give an idea of the random fluctuations in the value due to actual experience being different from the assumptions made in the basis.

[1]

Further, the method and basis need to be discussed with the actuary of company Y and a consensus needs to be reached upon before proceeding with the assessment of liabilities of scheme Y.

[0.50]

[Max 7]

vi)

a) The Parties connected with the merger of scheme Y with scheme X are:

- Buyer company X,
- Seller company Y,
- Trustees of scheme X,
- Trustees of scheme Y,
- Employees of company X who are members of scheme X and who are members of DC scheme and the unions representing them, if any
- Employees of company Y and the unions representing them,
- Actuaries, administrators & legal advisors of both the companies,
- Income tax authorities etc.

Concerns & conflicting interest of different stake holders may be as follows:

Trustees of scheme X will have concern for:

- Funding level of scheme Y which is quite low compared to their scheme X and will affect the security of benefits of their members as the funding level of combined scheme will reduce unless deficit is funded either by company Y before merger or after Company X after merger
- Even if the deficit of scheme Y is funded on realistic basis, there may remain deficit on current valuation basis of scheme X and hence trustees of scheme x may like for additional contribution from Company X after merger in order to

[0.50]

[0.50]

protect security of benefits of combined scheme

- The eligibility conditions and benefits of scheme Y are quite different from scheme X and hence will create problems of administration of combined scheme [0.50]

Company X would be interested in removal of deficit of scheme Y before merger so that they need not be required to fund the same after merger. In case company Y is not able to fund the deficit of scheme Y before merger, Company X will like the deficit to be adjusted in the purchase price to be paid to company Y. [1]

Company X may like to offer benefits of scheme X at least prospectively to the employees of company Y on merger, if the same are lower than scheme Y. They may also like to offer DC scheme [0.50]

Company Y will be interested:

- That they are not required to pay the deficit of scheme Y, if any before the deal is finalized nor they would like the deficit be adjusted in purchase price [0.50]
- The benefits available to its employees in scheme Y are not reduced after merger, particularly the accrued benefits [0.50]

Trustees of scheme Y will also be interested in protecting the interests of their members, i.e. they get better deal in respect of benefits and security in the combined scheme on merger [0.50]

Employees of Company X (and their employee unions, if any) who are members of scheme X will have concern if eligibility/vesting conditions and/or benefits of scheme Y are more favourable compared to their scheme X, particularly if the funding level of scheme Y remains lower than 100%. [0.50]

Similar will be the case for employees of company Y and their unions if scheme X benefits are more favourable than scheme Y. They would like their benefits are not reduced on merger. Rather, they may insist for benefits of scheme X on merger, if they are more favourable. [0.50]

Employees of Company X who are members of the DC scheme (and their unions, if any) may have concern if they are not satisfied with DC scheme and want to be part of DB scheme. They may raise a demand to be part of DB scheme on merger on the plea that some employees whose joining date is subsequent to them in company Y will be enjoying DB scheme benefits whereas they will not. [1]

Actuaries of both the companies will be interested in striking an agreeable basis for valuation of liabilities of scheme Y, though each will try to favour their client- actuary of company X will be interested in arriving at a higher liability whereas the actuary of scheme Y will try to arrive at a lower liability. [0.50]

Income tax authorities will be interested that purchase price of company Y is not adjusted unduly for deficit of scheme Y to lower the tax liability, if any [0.50]

[8 Marks]

- b)** Since the benefits & other eligibility conditions of both the schemes X & Y are quite different, Company X can run two different schemes on merger. Both these schemes [0.25]

will be closed to new entrants as new employees of Company X will join its DC scheme

Transfer value may be calculated for each member of scheme Y for their accrued benefits on date of joining into Company X as an initial contribution and the members may be allowed to join DC scheme of Company X effected through NPS.

[0.50]

If most of the employees of company Y don't join Company X or are not allowed to join the Company X, such employees may prefer to have cash value of their accrued benefits in scheme Y. Such cash value may either be taken by them net of tax or got transferred to the pension scheme of their new employer. Remaining small number of employees of Company Y may either be offered benefits in scheme X or cash value of accrued benefits may be accepted to DC scheme of Company X.

[0.50]

Company/Trustees of scheme Y may buy out accrued benefits of employees from insurance companies to discharge their pre-merger pension liabilities. The employees of Company Y who join the Company X on merger may be offered benefits under DC scheme.

[0.50]

Company Y may run the scheme Y for accrued benefits with future salary increases on an agreed basis. The employees of company Y who join company X may be offered to join DC scheme

[0.50]

[Max 2]**[50 Marks]**

Solution 2:

- i) Stakeholders - employer, employee, trustees, government, beneficiaries, taxation authorities [0.25]
- Current wage definition - varies across industries and companies [0.25]
- Government:** Proposed wage definition - aimed at bringing about consistency in wage definitions [0.25]
- To ensure that companies do not structure wages in a manner that minimizes the contribution to Retirement Funds; aim to strengthen the pension system in the country for organized sector [0.50]
- Would require for details on how these rules be applicable with examples [0.25]
- With specific reference to treatment of allowances that are paid to all uniformly, reimbursement of expenses, commissions and components that are variable OR based on the employee performance like incentives [0.50]
- There could be other aspects relating to tax benefits for contributions and benefit payments that need to be looked into from the perspective of maximum allowable limits [0.50]
- Employer:** While the intent is good, this new wage definition would imply a cost impact on P&L both for Provident Fund and Gratuity [0.25]
- The new wage definition would increase the wages used for calculating contribution towards PF. So company cost would increase unless contribution to PF is a part of cost to company (CTC) [0.50]
- However, the impact may not be very significant if the company restricts the salary for PF contribution to K 20,000 [0.25]
- The impact on Gratuity would be expected to be more than that for Provident Fund [0.25]
- Given that it is a final salary defined benefit scheme [0.25]
- The materiality of the impact would depend upon the current wage structure [0.25]

For Gratuity benefit it will also depend upon the demographic profile of the company	[0.25]
Higher and more material for older organisations with tenured employees as compared to younger organisations.	[0.50]
and whether the benefit is as per the Act or more generous	[0.25]
If the benefit is as per the Act, the pace at which the benefit would get closer to the maximum payable would increase so liability would get closer to the current point in time	[0.50]
It will not impact the liability for those who have already reached the limit	[0.25]
However, whenever the cap of gratuity is enhanced by the government in future, there will be immediate impact on liability	[0.25]
If the benefit is more generous than the Act, the liability in respect of each employee would increase and hence overall liability would also increase	[0.50]
Companies may restructure their salaries to minimise impact	[0.25]
There may be changes required in payroll and employees need to be communicated about the new salary structure	[0.25]
Employees: Matching contribution by employee for Provident Fund implies take home salary would be expected to reduce	[0.25]
However, the retirement benefit will increase due to enhancement of contribution	[0.25]
Also the move of the government will increase tax efficient retirement savings	[0.25]
Further, the impact may not be there if the employee draws a higher salary and if the company restricts the contribution to a salary of K 20,000	[0.25]
If the Gratuity benefit has no ceiling, the employee would get a higher benefit as compared to before	[0.25]
However, it would be subject to tax beyond K 2,500,000	[0.25]
The older employees who are closer to retirement may benefit much more as compared to younger employees due to past service benefit going up as it is a final salary benefit	[0.50]
Trustees:	
If the gratuity scheme is funded under a trust, then the funding ratio will drop immediately on implementation of the move as past service liability will go up	[0.50]
The trustees would demand for additional contribution from the employer in order to bring funding level to the desired level	[0.25]
This will be in spite of the regular contribution going up	[0.25]
Beneficiaries:	
The death benefit available for beneficiaries under gratuity will enhance immediately due to increase in accrued gratuity	[0.25]
Under PF it will increase over time due to increase in contribution	[0.25]
Taxation Authorities:	
The tax collection is likely to be affected adversely which will be a concern of tax authorities. It is due to profit of companies may go down due to which corporate tax collection may be affected	[0.50]
Also individual tax collection may go down due to increase in tax efficient retirement benefit savings	[0.25]
Change in take home salary may impact the consumption pattern of salaried class and impact the economy	[0.25]
[Max 11]	

- ii) Gratuity is a defined benefit scheme and so the cost of providing benefits cannot be ascertained precisely until all benefits are paid and there is no member in this scheme [0.50]

	Costs can be estimated in advance based on expected cash flows determined on the basis of assumptions relating to future salary growth, employee turnover, mortality etc.	[1]
	However, the ultimate cost of the scheme would depend upon how the actual experience is borne out in practice	[0.50]
	As per IND AS19, the cost to be recognised is based on accrual principle	[0.25]
	The cost is recognised when the benefit is earned and not when the benefit is paid and hence enhanced cost will impact the financials of the company immediately	[0.50]
	Since it is a final salary scheme, the gratuity payable to employees would be increased effectively for all years of past service, even if the new definition of wages would be effective at a future date of 1st July 2021	[0.50]
	The impact on an entity depends upon the current wage structure, demographic profile of the company, tenure of the employees and the maximum benefit payable	[0.50]
	Since Entity A has a wage structure that is currently 30% of the Total Cash as basic the impact would be material when the company restructures the wages to 50% of Total Cash	[0.50]
	Further, since there is no ceiling on maximum benefit payable, it will impact accrued benefit of all employees and more of tenured employees with higher salaries	[0.50]
	Entity B, on the other hand, is currently at 45% of the Total Cash as basic so the impact would not be as material as that for Entity A	[0.50]
	Further, since the benefit is capped at K 2,500,000, there would be no impact on benefit of employees who have already hit the maximum benefit payable.	[0.50]
	Further, as per the Accounting Standards, any change in the present value of the defined benefit obligation for employee service in prior periods, resulting from a plan amendment is to be treated as a Past Service Cost	[1]
	A Plan amendment is the introduction or withdrawal of, or changes to, a defined benefit plan	[0.50]
	Further, as per clause 89 of IND AS 19, any future benefit changes that are not set out in the formal terms of the plan (or a constructive obligation) at the end of the reporting period are not reflected through actuarial assumptions. Such changes will result in past service cost, to the extent that they change benefits for service before the change	[2]
	Para 102 to 108 of IND AS 19 mention the conditions under which Past Service Cost would apply	[0.50]
	If the company does not restructure wages and simply includes the new components while calculating the Gratuity liability, it may be treated as plan amendment	[0.50]
	So there would be a Past Service Cost associated that needs to be recognised immediately through Income statement	[0.50]
	If the company restructures wages, particularly before the government notification then the increase in liability may be treated as actuarial gains/losses and would be routed through Other Comprehensive Income (OCI)	[1]
		[Max 11]
iii)	Assumptions for actuarial valuations are long term in nature	[0.25]
	Clause 90 of IND AS 19 states that estimates of future salary increases take account of inflation, seniority, promotion and other relevant factors, such as supply and demand in the employment market	[0.50]
	The Covid 19 situation is not expected to stay in the long term. There may be freeze on salary increases to manage the costs in the short term, but on a long term basis the company would need to give increases to address relating to promotion, demand and supply etc.	[0.50]
	Further, the inflationary effect has also to be considered in the salary escalation assumption in the long term	[0.25]

In future, when salaries would increase, the gratuity liability will go up affecting the financials of the company	[0.25]
Moreover, the liability will also increase due to increase in salary escalation assumption rate	[0.25]
	[Max 2]

iv) Advantages for employers/employees:

Increased security for employees as the defined benefit scheme assets are held separate from the employer's business.	[0.25]
Reduced volatility of expenses on an accounting basis	[0.25]
Better liquidity management and future cashflows	[0.25]
Tax incentives to company for contributions	[0.25]
Aligned to global best practices for such benefits	[0.25]
Benefit administration can be outsourced to fund manager if funds managed by an external fund manager	[0.25]

Disadvantages to employers

Opportunity cost	[0.25]
More oversight on plan governance for trust related matters	[0.25]
Management charges by fund manager if fund management is outsourced	[0.25]
Exposure to investment risks in respect of the fund	[0.25]
Exposure to over funding - Typically there may be legislation that restrict the level of funding allowed in a scheme in order to qualify for tax concessions.	[0.50]
Exposure to regulatory compliances for trust	[0.25]

Trustees:

A trust fund is created separately under a trust for payment of gratuity/other defined benefit to the eligible members/beneficiaries	[0.25]
Trustees act as per Trust Deed and Rules of the Scheme	[0.25]
They can take help of specialists depending on the size of the scheme	[0.25]

Government/Tax Authorities:

Government may like to encourage funding of retirement savings so that funds are available in the system for long term projects	[0.25]
Though there will be lower collection of corporate tax if tax advantages are associated with funding of retirement benefits	[0.25]

Regulator:

Though there is no mandatory funding requirement at present, the Regulator may be needed to have a check on overfunding for tax avoidances	[0.50]
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Insurance Companies:

Life insurers help in funding by introduction of specific plans which may be linked/non-linked.	[0.25]
Funding through insurers may be more advantageous for small employers	[0.25]
Sometimes life insurers offer attractive life cover due to which gratuity/defined benefit available on death is higher though they will charge additionally for it	[0.25]

Other Specialists such as Fund Managers/Actuaries/Legal Advisers:

Services of other specialists may help trustees whenever they need any advice and/or to discharge their duties, say fund management, administration etc.	[0.25]
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[Max 6]

v) In a defined benefit scheme the benefits are defined in advance and the cost of providing

	those benefits is unknown	
	The contributions may be needed from the company depending on the level and form of benefits of the scheme	
	The company may fund the scheme in which case it may pay contributions during the period when they accrue or the company can have unfunded scheme and pay contributions when the benefits fall due for payment	[0.50]
	The company covenant for such defined benefit scheme is the:	
	Ability and	[0.25]
	Willingness	[0.25]
	to pay sufficient contributions	[0.25]
	to meet the benefits as they fall due	[0.25]
	Trustees should assess the sponsor's covenant unless	
	The scheme is well-funded, therefore, no reliance is placed on sponsor	[0.25]
	The sponsor covenant is strong enough to be deemed as certain	[0.25]
	The sponsor covenant is so weak as to be deemed nil	[0.25]
	The sponsor has no further liability	[0.25]
		[Max 2]
vi)	Depends upon current funding level of scheme	[0.25]
	If the benefit is funded, other things remaining the same, the funded position would be affected adversely as the accrued liabilities of the scheme would increase i.e. funding level would fall. Companies would need to increase the contributions to reach the desired level of funding	[0.50]
	This would further put pressure on the immediate cash flow for the employer	[0.25]
	The funding level will be adversely affected irrespective of whether the employer restructures its salary structure or not	[0.25]
	However, if the contributions are treated as a deductible business expenses for the company then given that the overall wage bill would increase, the company may get additional tax benefits	[0.50]
	Companies may also decide to go slow on funding as the country does not have a minimum funding level requirement	[0.25]
		[3]
vii)	Depends upon the current regulations and laws in the country	[0.25]
	Typically tax laws make it mandatory to set up a Trust and obtain relevant approvals/recognitions from the regulator/revenue authorities for the contributions to be permissible as deductible expenses.	[0.50]
	Subscribing to a scheme from an insurance company directly may not be an alternative to the Trust.	[0.25]
	Segregation of assets / liabilities - A Trust ensures clear segregation of assets and liabilities due to being an independent legal entity distinct from the company.	[0.25]
	It ensures complete accounting of the fund and safeguards from any unauthorized access to the fund other than the purpose for which the Trust has been set up.	[0.25]
	Investment Governance: A recognized Trust whose funds are managed by an insurer provides greater investment governance.	[0.25]
	Security selection and other tactical actions are the preserve of the insurer while the Trustees focus on oversight and review of performance.	[0.25]
	The trust is typically irrevocable and cannot be wound up till the last benefit payment has been made, subject to the availability of funds. In that sense no moneys belonging to the Fund in the hands of the Trustees shall be recoverable by the Employer	[0.50]

Purchase of an insurance policy is considered to be an investment choice and cannot replace the benefits of the Trust [0.25]

By purchasing an insurance policy, the employer remains the owner of the policy and hence will have control on the policy. It therefore defeats the very purpose of providing security of employee benefits by funding them [0.50]

[Max 3]

viii) Self- managed

Provides greater flexibility in investment of funds [0.25]

Subject to regulatory restrictions on investment of retirement funds [0.25]

Would need to manage investments, administration etc. in-house [0.25]

Can use different providers for different services, e.g. an administration, investments, actuarial specialist, depending upon the size of the scheme [0.25]

Need to regularly monitor the portfolio to ensure compliance with investment regulation [0.25]

May not achieve diversification of investments to the extent an insurer can have by managing the funds of several schemes [0.25]

Funding with an insurance company [0.25]

Typically would have an investment product wherein the choice of assets is determined by the insurance company [0.25]

The product could be a traditional product or unit linked [0.25]

In case of unit linked products, there could be choice of investments. Further, the expenses could be explicit charges, e.g. bid/offer spread, annual management charge, an annual policy fee and a separate administration fee. [0.50]

Or could be implicit charges, e.g. returns declared net of expenses. [0.25]

A min guaranteed return may be mandated by the regulator [0.25]

The insurance company may also be able to provide other services such as benefits administration, actuarial services, funding advice etc. [0.25]

Though there may be a conflict of interest for the fund manager and the employer may need to establish independence [0.25]

There could be additional insurance benefits that may be bundled into the product [0.25]

This would be advantageous for companies that have either a small membership or a low value of assets [0.25]

Even if funds are managed by an insurance company, the trustees would continue to have a fiduciary responsibility towards the members [0.50]

The trustees would need to monitor return of the overall fund, including performance evaluation vis à vis benchmarks and peers [0.50]

[Max 5]

ix) Strategy with specific reference to liabilities [0.25]

Objective may be to hedge inflation and interest rate risks [0.25]

The discount rate used for actuarial valuation of liabilities for accounting of employee benefit costs purpose is required to be driven by market-based rate of interest through government or corporate bond yields. So the estimated value of liabilities is exposed to interest rate risks [0.50]

However, interest rates and changes to interest rates have no impact upon the actual amount paid out [0.50]

While the future anticipated inflation assumption is expected to be built into salary growth rate, the scheme is still exposed to inflation risk as the actual inflation may turn out to be different from that expected [0.50]

Given the long duration of liabilities, the present value of defined benefit scheme liabilities is extremely sensitive to movements in inflation rates. [0.25]

Liabilities of defined benefit schemes typically have a longer duration as compared to duration of assets	[0.25]
So there is exposure to interest rate and reinvestment risks	[0.25]
Exact matching of cash-flows may be very difficult given the non-availability of assets of desired duration	[0.25]
Even if exact matching of assets with estimated liabilities is achieved, the actual experience may change over time, say salaries increase more than expected, attrition may be more or less, sudden raising of benefit cap by the government etc. In that case the assets need to be relooked into	[0.50]
Hedging implies that the assets should move in same direction as liabilities due to inflation or interest rate change	[0.25]
Bonds would perhaps be the closest to address this aspect	[0.25]
Index linked bonds or swaps could be used (if available and allowed by regulation)	[0.50]
Swaps would expose the scheme to counter party risks	[0.50]
In the absence of index linked bonds, equity exposure should help hedge the inflation risk	[0.25]
Since the gains/(losses) from return on investment are recognised through OCI, this should encourage trustees to look at long term real assets	[2]
A target return approach can also be used to match or beat the discount rate	[0.50]
It would of course depend upon the regulation for investment of defined benefit scheme funds	[0.25]
There would be no incentive to generate investment surplus to manage future contributions.	[0.25]

[Max 8]**[50 Marks]**
